

Vulnerability of Heterogeneous Graph Learning Models to White-Box Feature Attacks

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Abstract—Graph neural networks (GNNs) yield highly effective results in security-oriented node classification by incorporating both node features and inter-node relationships into the learning process. However, the adversarial robustness of these models remains a critical concern. This study models DNS-based client-domain interactions as a heterogeneous graph structure to classify client nodes as malicious or benign. We developed and evaluated four heterogeneous graph learning models (HeteroSAGE, HeteroGCN, HeteroGAT, and HeteroGAE) using clean test data, followed by three white-box adversarial attacks (FGSM, PGD, and BIM) targeting node features. Performance was measured using accuracy, precision, recall, F1-score, and attack success rate. Results show that while HeteroGAT demonstrates balanced performance and stability under attack, HeteroSAGE is highly susceptible to adversarial perturbations despite high baseline accuracy. These findings highlight that evaluating security-focused graph models requires assessing their resilience to adversarial attacks rather than relying solely on clean-data performance.

Index Terms—Graph Neural Networks, Heterogeneous Graph Learning, Adversarial Attack, FGSM, PGD, BIM

I. INTRODUCTION

Classical deep learning has achieved remarkable success in numerous fields, such as computer vision, natural language processing, and speech recognition. However, while these conventional approaches utilize the individual features of observed entities, they often ignore the relationships among them during the learning process. To address this limitation, graph neural network (GNN)-based approaches leverage both node features and inter-entity relationships, enabling them to model complex dependency patterns that traditional models fail to capture [1], [2]. GNNs have recently gained widespread popularity across various relational data domains, including social networks [3], knowledge graphs [4], recommendation systems [5], and security logs [6]. This adoption is primarily driven by the capacity of GNN architectures to integrate not only node-level features but also the topological dependencies between entities into the model.

Despite the growing ubiquity of traditional deep learning methods, adversarial attacks targeting these systems present a significant security threat. These attacks involve carefully designed, imperceptible perturbations applied to the input data. Although these deliberate alterations keep the inputs visually or statistically indistinguishable from the original data, they can successfully manipulate model predictions. Pioneering works such as the Fast Gradient Sign Method (FGSM) [7], Basic Iterative Method (BIM) [8], and Projected Gradient Descent (PGD) [9] have demonstrated that even highly accurate vision models remain susceptible to such perturbations.

Adversarial attacks are also applicable to graph-based learning methods. However, systematic studies investigating their impact remain limited. Most existing research focuses on manipulating discrete graph structures, such as edge addition or deletion. Furthermore, prior literature has examined adversarial attacks on models only individually. To our knowledge, no study compares common heterogeneous graph learning architectures within a unified experimental framework. Consequently, these models have not been evaluated under identical data splits, attack conditions, and evaluation metrics.

This study systematically investigates the impact of adversarial attacks targeting graph-learning methods on model performance. Specifically, three widely used gradient-based attack methods (FGSM, PGD, and BIM) are applied to four distinct graph neural network architectures to evaluate model response at various perturbation intensities. Our work focuses on a node classification scenario, exploring how perturbations engineered in the feature space affect the classification accuracy of graph-based models.

The remainder of this paper is organized as follows. Section 2 summarizes the related work and existing literature in this domain. Section 3 provides the foundational background theory, detailing graph-based learning, graph neural networks, adversarial attacks, and the specific graph learning architectures analyzed in this study. Section 4 outlines the experimental design, describing the dataset, the environment, and the setup procedures. The empirical findings are presented in Section

5, followed by a comprehensive evaluation, discussion of the results, and concluding remarks in Section 6.

II. RELATED WORK

Adversarial attacks were first systematically investigated in the domain of image classification. Proposed by Goodfellow et al. [7], the Fast Gradient Sign Method (FGSM) is recognized as a foundational method that generates adversarial examples in a single step by utilizing the sign of the gradient of the loss with respect to the input. Kurakin et al. [8] extended this approach into a multi-step framework to propose the Basic Iterative Method (BIM) attack. Similarly, Madry et al. [9] established the Projected Gradient Descent (PGD) attack as a primary reference for robustness analysis and adversarial training. Finally, Han Xu et al. [10] comprehensively summarizes the roles of classic attacks across diverse data modalities, including images, graphs, and text.

With the widespread adoption of graph neural networks (GNNs), the vulnerability of these models to adversarial attacks has also become a major focus of research. Proposed by Daniel Zügner et al. [11], Nettack stands out as one of the first systematic attacks targeting node classification models on attributed graphs. Nettack demonstrated that target nodes can be misclassified under both evasion and poisoning attacks through a limited number of edge additions/deletions and attribute modifications. Additionally, this method ensures that perturbations remain unnoticeable without significantly altering global dataset statistics, making them difficult to detect.

While early adversarial studies on graphs primarily focused on structural (edge) manipulation, more recent research demonstrates that directly targeting node attributes can be just as effective as edge perturbations. Xu et al. [12] showed that node attributes can be directly targeted in both evasion and poisoning scenarios. Furthermore, they revealed that the GNN training process can be significantly degraded by incorrect or deliberately corrupted labels, showing that model behavior can be manipulated solely through trigger patterns without altering the labels themselves.

Similarly, Zheng et al. [13] demonstrates that both attribute manipulations and structural perturbations severely weaken performance in GNN-based anomaly detection, and they propose various robustness-enhancing mechanisms to counter these threats. In their study on label poisoning attacks, Liu et al. [14] illustrated that the GNN training process can be disrupted using fake or modified labels, utilizing gradient-based maximum perturbation strategies. Other studies have designed backdoor attacks utilizing node attributes, reporting that small attribute patterns can be used to covertly control model behavior [15].

Previous studies have demonstrated the vulnerability of GNN models to structural perturbations, node feature alterations, label poisoning, and other adversarial threats; however, these works utilize varying graph types and attack assumptions. This study investigates the adaptation of classical white-box adversarial attacks to graph-based learning methods via

node attributes. By maintaining fixed structural connectivity and modifying only node features, this research comparatively analyzes the robustness of four heterogeneous architectures against a unified detection task using standardized performance metrics.

III. BACKGROUND

This section details the fundamental technologies utilized in this study: graph-based learning, graph neural networks, and adversarial attacks. First, the underlying framework of graph-based learning is explained, followed by an analysis of the operational principles of graph neural networks. Finally, the adversarial attack approaches employed to evaluate the robustness of these models are presented.

A. Graph-Based Learning

Unlike traditional machine learning, graph-based methods leverage a relational inductive bias, making predictions for a given node depend on both its own attributes and the features and relationships of its neighbors. This approach is crucial in domains with explicit graph-structured dependencies, such as fraud detection, malware propagation, and DNS-based malicious domain detection.

The problem is formulated over an attributed graph $G = (V, E, X)$, where V is the set of N nodes, E is the set of edges, and X represents the node attribute matrix, with x_i denoting the attribute vector for node $v_i \in V$. Graph connectivity is represented by the adjacency matrix A . In node classification, given a labeled subset $V_L \subset V$, the objective is to learn a mapping f to predict labels for all nodes by utilizing both X and A , thereby integrating individual node features with structural dependencies.

B. Graph Neural Networks

Graph Neural Networks (GNNs) perform relational inference via a message-passing mechanism, updating node representations by iteratively aggregating neighbor information. A general L -layer GNN is formulated as:

$$h_i^{(0)} = x_i, \quad (1)$$

$$h_i^{(\ell+1)} = \text{UPDATE}^{(\ell)}(h_i^{(\ell)}, M_i^{(\ell)}), \quad (2)$$

$$M_i^{(\ell)} = \text{AGGREGATE}^{(\ell)}(\{h_j^{(\ell)} : j \in \mathcal{N}(i)\}). \quad (3)$$

where $\ell = 0, \dots, L-1$ is the layer index and $\mathcal{N}(i)$ is the neighborhood of node i . The final layer representation, $h_i^{(L)}$, is input to a classifier for node-level prediction.

GNN variants differ in their implementation of the AGGREGATE and UPDATE operators. For example, Graph Convolutional Networks (GCNs) serve as a robust baseline by aggregating normalized neighbor features. GraphSAGE utilizes neighborhood sampling and aggregation to handle large-scale and inductive tasks. Graph Attention Networks (GATs) employ attention mechanisms to weight neighbor contributions dynamically. Finally, Graph Autoencoders (GAEs) reconstruct graph structures via an encoder-decoder architecture, making

them highly effective for representation learning, link prediction, and uncovering latent structural patterns in label-scarce scenarios.

C. Adversarial Attacks

The adversarial attacks evaluated in this study are white-box attacks, assuming an adversary with complete access to model parameters and input loss gradients. Given a trained classifier f and input x with ground-truth label y , the adversary seeks a perturbed input x' that causes misclassification by f , while constraining the perturbation δ under a selected norm (e.g., L_∞). Prominent established white-box methods include the Fast Gradient Sign Method (FGSM), Basic Iterative Method (BIM), and Projected Gradient Descent (PGD).

1) *FGSM (Fast Gradient Sign Method)*: The Fast Gradient Sign Method (FGSM) is a highly efficient, single-step attack that maximizes the loss by perturbing the input in the direction of the gradient sign [7]. Goodfellow et al. define the perturbation as $\eta = \epsilon \text{sign}(\nabla_x J(\theta, x, y))$, leveraging backpropagation to distribute the total perturbation budget along the most adversarial direction for each input dimension [7]. This yields the adversarial input:

$$x' = x + \epsilon \text{sign}(\nabla_x J(\theta, x, y)). \quad (4)$$

Although computationally efficient, FGSM is generally less robust than iterative attacks due to its single-step nature. Consequently, it is primarily characterized in the literature as a low-cost generator suited for rapid initialization analysis [7], [9].

2) *BIM (Basic Iterative Method)*: The Basic Iterative Method (BIM) is an iterative variant of FGSM that executes multiple small steps. After each step, a clipping function, $\text{Clip}_{x,\epsilon}(\cdot)$, constrains the intermediate representation within the ϵ -neighborhood of the original sample x [8]. The update rule is formulated as [8]:

$$x_0^{\text{adv}} = x, \quad (5)$$

$$x_{t+1}^{\text{adv}} = \text{Clip}_{x,\epsilon}(x_t^{\text{adv}} + \alpha \text{sign}(\nabla_x J(\theta, x_t^{\text{adv}}, y))). \quad (6)$$

where α denotes the step size. This formulation incrementally increases the loss at each iteration while respecting the total perturbation budget.

By iteratively tracing the loss landscape rather than relying on a single-step approximation, BIM is typically more potent than FGSM. However, this comes at the expense of higher computational complexity, making the selection of both the step size α and the number of iterations highly critical [8].

3) *PGD (Projected Gradient Descent)*: Adversarial attack generation is formulated as an optimization problem where the multi-step Fast Gradient Sign Method (FGSM) is equivalent to projected gradient descent (PGD) on negative loss [9]. PGD updates in the direction of the gradient sign and projects back onto the feasible set. A key to PGD's empirical effectiveness is its random initialization within an ϵ -ball, which more reliably explores the inner maximization problem compared to the similar Basic Iterative Method (BIM) [9]. Consequently,

TABLE I
ATTRIBUTES OF CLIENT AND DOMAIN NODES

Node Type	Attribute Name (Variable)	Description
Client Nodes	Total DNS requests (count_dns_requests)	Total number of DNS queries generated by the client.
	Distinct DNS requests (count_distinct_dns_requests)	Number of distinct DNS queries generated by the client.
	Average requests per minute (avg_req_per_min)	Average number of DNS queries generated per minute.
	High request intensity minutes (high_req_per_min)	Number of minutes with high DNS query intensity.
	Successful DNS responses (count_response_success)	Total number of successful DNS responses received.
	Failed DNS responses (count_response_failed)	Total number of failed DNS responses received.
Domain Nodes	Unique querying clients (unique_client_queried)	Number of unique clients that queried the domain.
	Maliciousness score (malicious_score)	Score indicating the likelihood that the domain is malicious.

PGD is a strong first-order attack characterized by a general projection framework and comprehensive local exploration.

IV. EXPERIMENT

This section details the dataset and experimental setup phases utilized to evaluate the behavior of graph-learning-based models under adversarial attacks.

A. Dataset and Experimental Environment

This study utilizes the TI-2016 dataset of DNS client-domain interactions. Raw records were aggregated into a heterogeneous graph with client and domain nodes linked by directed query edges, alongside reverse edges to facilitate bidirectional message passing [16]–[18]. Graph topology was derived from categorical identifiers, while numerical attributes provided node representations. Detailed graph construction and feature extraction follow [18]. Six client-node and two domain-node features are summarized in Table I.

Before graph construction, identifiers were standardized, and numerical features were scaled using training set statistics. Labeled client nodes were partitioned into training (70%), validation (15%), and test (15%) sets, preserving class distribution. Adversarial attacks targeted only client-node features; domain features and graph topology remained fixed.

Framed as a node classification task, the primary objective is identifying client nodes as malicious or benign. All models were evaluated under identical data and conditions to ensure a fair comparative analysis across architectures.

B. Experimental Setup

Four distinct graph-learning-based models were developed for the experimental evaluation: HeteroSAGE, HeteroGCN, HeteroGAT, and HeteroGAE [4], [19]–[22]. These models represent diverse learning paradigms, such as message passing over graphs, attention mechanisms, scalable aggregation,

TABLE II
SUMMARY OF HETEROGENEOUS GRAPH MODELS

Model	Base Architecture / Layer	Graph Structure	Hidden Dimensions	Regularization	Output / Loss Function
HeteroSAGE	SAGEConv-based message passing	Heterogeneous bidirectional client-domain graph	$(64, 32) \rightarrow (96, 64) \rightarrow (64, 48)$	Dropout = 0.3, Batch-Norm, $L2 = 10^{-4}$	Sigmoid, BCEWithLogits
HeteroGCN	GCNConv-based message passing	Heterogeneous bidirectional client-domain graph	$(64, 32) \rightarrow (96, 64) \rightarrow (64, 48)$	Dropout = 0.3, Batch-Norm, $L2 = 10^{-4}$	Sigmoid, BCEWithLogits
HeteroGAT	GATConv-based attention	Heterogeneous bidirectional client-domain graph	$(64, 32) \rightarrow (96, 64) \rightarrow (64, 48)$	Dropout = 0.3, attention dropout = 0.2, BatchNorm, $L2 = 10^{-4}$	Sigmoid, BCEWithLogits
HeteroGAE	GNN encoder + inner-product decoder	Heterogeneous bidirectional client-domain graph	$(64, 32) \rightarrow (96, 64) \rightarrow (64, 48)$	Dropout = 0.3, Batch-Norm, $L2 = 10^{-4}$	Sigmoid, BCEWithLogits

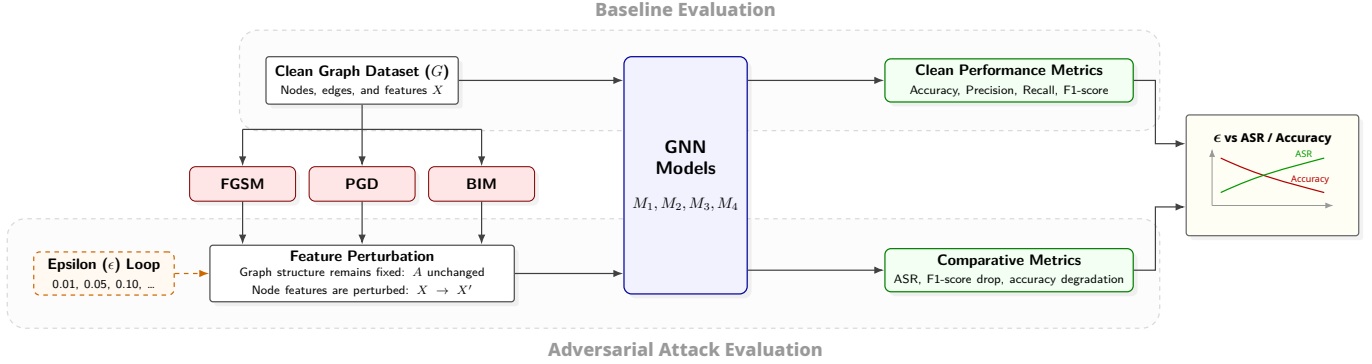


Fig. 1. Framework for evaluating the robustness of GNN models against feature-based adversarial attacks.

and representation learning. The architectural details of these models are presented in Table II.

As illustrated in Fig. 1, the models were initially evaluated on clean test data, after which FGSM, PGD, and BIM attacks were sequentially applied to each model. These attacks were executed directly on the node features while keeping the graph topology constant. This approach allowed for an investigation into the extent to which attacks influence model decisions through perturbations in the feature space, independent of any structural modifications.

Throughout the experiments, varying epsilon levels were employed to assess how increasing perturbation magnitude affects model performance. For each of the four models, twenty independent trials were executed under every combination of attack type and epsilon level, with the results reported based on the mean values of these runs.

C. Performance Metrics

To evaluate model performance on clean data and robustness against adversarial attacks, we utilized five metrics: accuracy, precision, recall, F1-score, and attack success rate (ASR).

While accuracy measures the proportion of correctly classified instances, potential class imbalances in the dataset make it insufficient on its own. Therefore, we incorporated precision and recall. Recall is especially critical in security contexts, as misclassifying a malicious client as normal severely compromises system security. Finally, the F1-score reflects both overall accuracy and balanced malicious class detection, serving as our primary metric to analyze post-attack performance degradation.

TABLE III
BASELINE PERFORMANCE VALUES

Model	Accuracy	Precision	Recall	F1-score
HeteroSAGE	97.00%	95.63%	98.51%	97.05%
HeteroGAE	97.19%	95.81%	98.71%	97.24%
HeteroGAT	98.15%	97.14%	99.23%	98.17%
HeteroGCN	86.67%	82.17%	93.67%	87.54%

$$ASR = \frac{\text{Successful Attacks}}{\text{Number of Target Nodes}} \quad (7)$$

To measure adversarial impact directly, ASR represents the proportion of target nodes, initially classified correctly on clean data, that are misclassified post-attack. Since the objective of our attacks is to induce the misclassification of malicious nodes as normal, ASR evaluates the extent to which model decisions are compromised.

V. RESULTS

On the clean test dataset (Table III), HeteroSAGE, HeteroGAE, and HeteroGAT achieve high accuracy and F1-scores, whereas HeteroGCN exhibits comparatively lower performance.

Fig. 2 shows that across all models, the Attack Success Rate (ASR) rises as the perturbation magnitude (ϵ) increases, indicating that growing feature perturbations increasingly disrupt graph-model decision boundaries.

As shown in Fig. 3, model F1-scores decline as the perturbation magnitude (ϵ) increases, though the degradation rate varies by architecture. HeteroSAGE's performance begins to decline

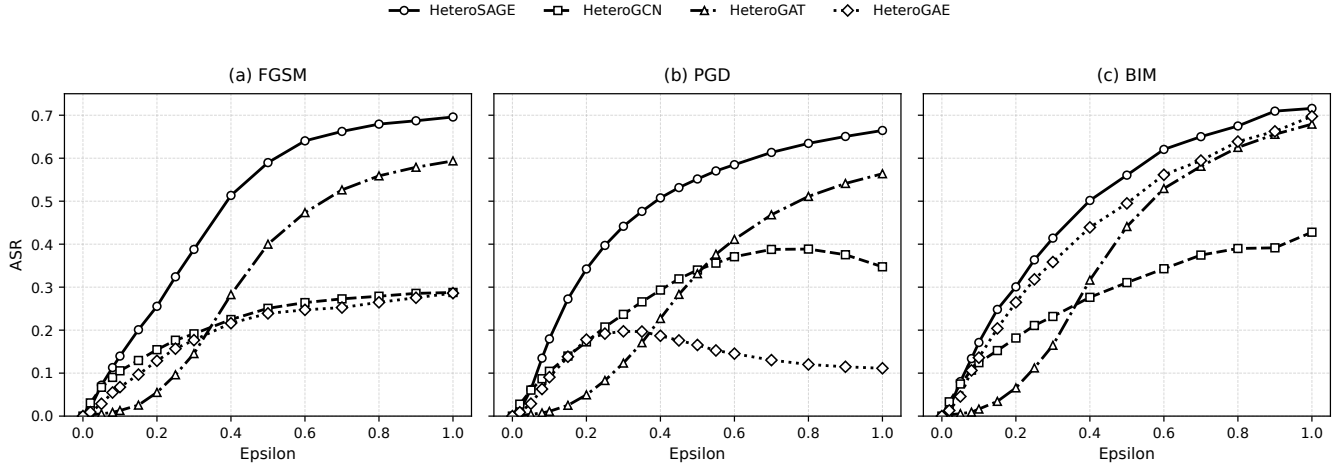


Fig. 2. Epsilon-ASR curves under FGSM, PGD, and BIM attacks.

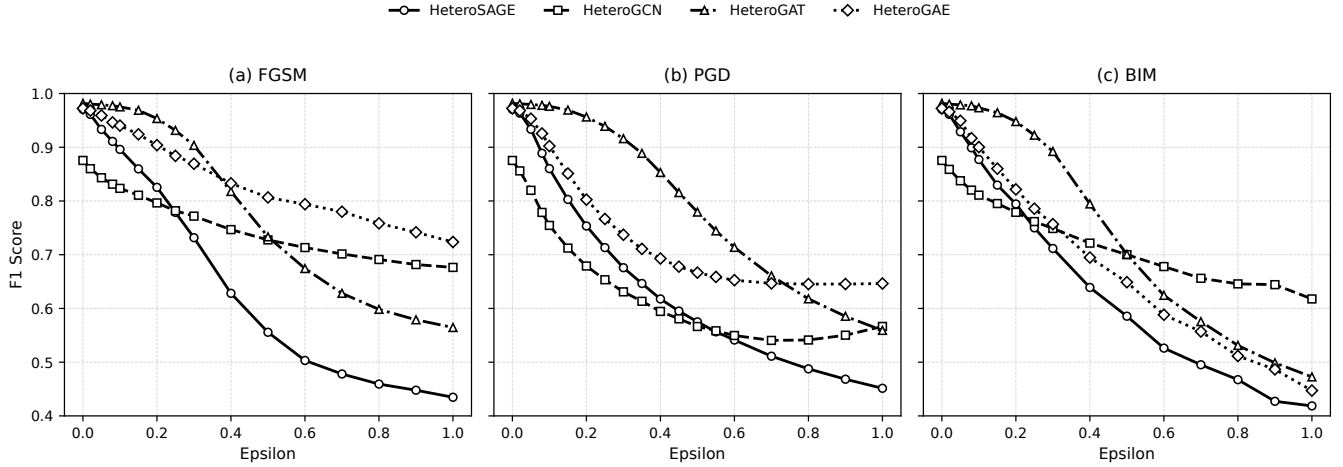


Fig. 3. Epsilon-F1 score curves under FGSM, PGD, and BIM attacks.

at lower ϵ values, whereas HeteroGAT exhibits a more gradual degradation. Meanwhile, due to its lower baseline, HeteroGCN yields the weakest absolute post-attack performance.

Across all models, accuracy declines as the perturbation magnitude (ϵ) increases (Fig. 4), with degradation trends depending on both the attack type and model architecture.

VI. DISCUSSION AND CONCLUSION

Evaluating HeteroSAGE, HeteroGCN, HeteroGAT, and HeteroGAE under feature-targeted, white-box attacks (FGSM, PGD, BIM) shows that increasing perturbation magnitudes (ϵ) degrades performance despite static graph structures. Architectural robustness varies: HeteroGAT balances clean accuracy and stability, HeteroSAGE remains vulnerable despite high clean performance, HeteroGCN’s lower attack success rate reflects low baseline accuracy, and HeteroGAE’s stability depends on attack type. Ultimately, no single architecture is consistently most robust across all attack types and perturbation levels.

Despite high baseline performance on clean data, HeteroSAGE degrades rapidly under adversarial attacks. This suggests that its GraphSAGE-based neighbor sampling and aggregation mechanism is unstable under feature-space perturbations, leading to sharp declines in F1-score and accuracy as ϵ increases.

HeteroGCN’s lower baseline performance complicates its post-attack evaluation. Its low Attack Success Rate (ASR) under certain configurations stems from poor initial classification accuracy—which reduces the pool of correctly classified nodes available to attack—rather than genuine robustness.

HeteroGAT is the most balanced model, combining strong clean-data performance with relative adversarial stability. Its attention mechanism uses learnable weights to aggregate neighbor information, limiting the impact of feature perturbations at low-to-moderate ϵ levels. However, its vulnerability at high ϵ levels shows that attention is not a standalone defense.

HeteroGAE’s performance varies by attack type. While its graph reconstruction objective yields robust clean represen-

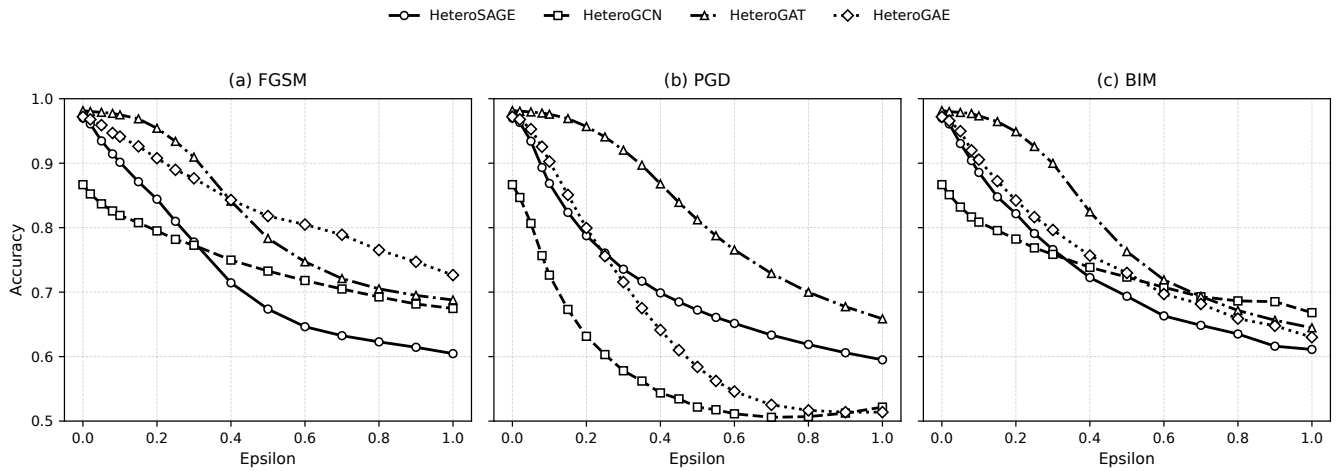


Fig. 4. Epsilon–Accuracy curves under FGSM, PGD, and BIM attacks.

tations, its severe degradation under iterative attacks (PGD, BIM) indicates that its latent space is highly sensitive to cumulative feature perturbations. Conversely, its resilience under one-step attacks (FGSM) underscores how the representation learning mechanism interacts with the attack type.

Collectively, rising ϵ degrades accuracy and F1-scores while increasing ASR across all architectures. This demonstrates that neighborhood relational structures alone cannot defend against feature perturbations; instead, model behavior is directly governed by specific message passing, attention, and representation learning choices.

Our findings demonstrate that high clean-data performance does not guarantee reliability, as top-performing models can degrade rapidly under escalating perturbation. Consequently, comprehensive evaluation must span diverse attack types and perturbation levels. Crucially, the attack success rate (ASR) must be interpreted alongside accuracy and F1-scores, as low ASR does not inherently signify robustness. Future research can enhance graph-based security by exploring structure-based attacks, joint feature-topology perturbations, adversarial training, and attack detection mechanisms.

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